Kinematics and integration

1

a $x(t) = 6t^2$

b $3\ \mathrm{s}$

2

a $s(t) = 3t^2 + 10t + 8$

b $133\ \mathrm{m}$

3

a $s(t) = 4t^3 + 15t^2 + 9t - 6$

b $914\ \mathrm{m}$

4

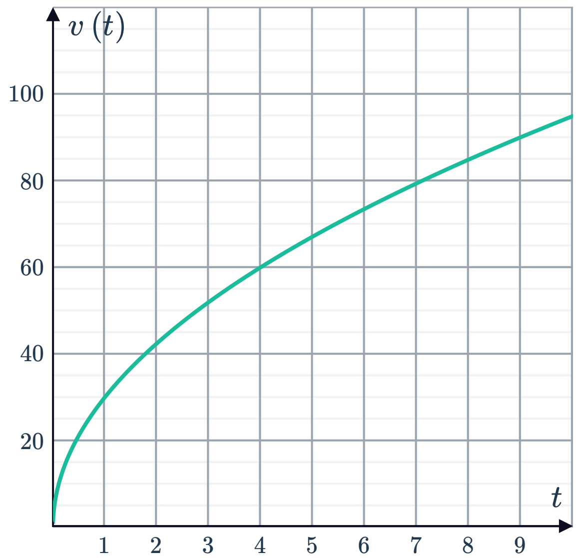
a $x(t) = 8\sqrt{t^3} + 5$

b $221\ \mathrm{m}$ to the right of the origin.

5

a $60\ \mathrm{m/s}$

b



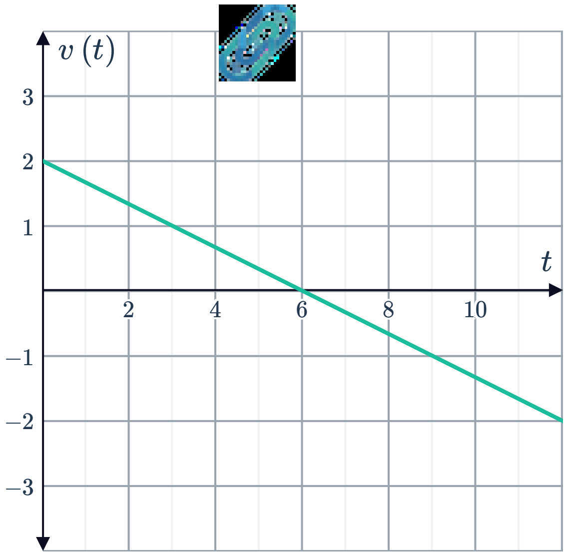
c $160\ \mathrm{units}^2$

d The distance travelled by the particle in the first 4 seconds. As the function is positive over this interval the area also represents the displacement of the object from its initial position after 4 seconds.

6

a $-1\ \mathrm{m/s}$

b



c 7.5 units^2

d The distance travelled by the particle in the first 9 seconds.

7

a 15 m/s

b $x(t) = 3t^2 + 15t + 3$

c 111 m

d 66 m

8

a 1 s

b $x(t) = 2t^2 - 4t + 3$

c

i 3 **ii** 1 **iii** 3

9

a 3 m/s

b $x(t) = 2t^3 - 7t^2 + 3t + 2$

c 30 m

d 34 m

10

a $x(t) = 4t^3 - 24t^2 + 5$

b $t = 0, 4 \text{ s}$

c -123 m

11

a -2 m/s^2

b $x = \dfrac{6t^{\frac{5}{2}}}{5} - t^2$

c $t = \dfrac{4}{9} \text{ s}$

d 2.87 m

12

a $x(t) = t^4 - 12t^3 + 36t^2 + 5$

b A turning point. At $t = 3$ the pen changes direction and starts moving to the left.

c 1.96 mg

13
a $5\left(4+e^{-5}\right)$ m
b 4.007 m
c 4.007 m
d Yes, because the velocity is always positive for $t \geq 0$.

14
a $v(t)=t^2-15t+56$
b $t=8, 7$ s

15
a $v(t)=3t^2-27t+60$
b $t=4, 5$ s

16
a $v(t)=2t-5$
b $x(t)=t^2-5t+8$
c 22 m to the right of the origin.
d $t=4$ s

17
a $t=2$ s
b 50 m
c $t=2+\sqrt{10}$ s
d $-10\sqrt{10}$ m/s

18
a $t=\dfrac{75}{49}$ s
b 13.5m
c $t=3.2$s
d -16.4 m/s

19
a $v(t)=-\dfrac{1}{4}e^{4t}+\dfrac{1}{4}$
b $-30\,322\,823$ cm

20
a 2.98 m/s
b 11.02 m

21
a 278 m/s
b $36\,203$ m

Trigonometric applications

22
a -18 m/s^2
b 3 m

23

a $-9\sqrt{3}\ \mathrm{m/s^2}$

b $3\ \mathrm{m}$

24

a $\dfrac{18}{\pi}\ \mathrm{m}$

b $\dfrac{30}{\pi}\ \mathrm{m}$

25

a $\dfrac{12\sqrt{2}}{\pi}\ \mathrm{m}$

b $\dfrac{96}{\pi}\ \mathrm{m}$

26

a $v\left(t\right)=-16\cos 4t+18$

b $26\ \mathrm{m/s}$

c $x\left(t\right)=-4\sin 4t+18t+3$

d $\left(2\sqrt{3}+6\pi+3\right)\ \mathrm{m}$

27

a $v\left(t\right)=10\sin 5t+6$

b $11\ \mathrm{m/s}$

c $x\left(t\right)=-2\cos 5t+6t+5$

d $\left(\sqrt{3}+\pi+5\right)\ \mathrm{m}$

28

a $0\ \mathrm{m/s}$

b $\dfrac{45}{\pi}\ \mathrm{m/s}$

29

a $0\ \mathrm{m/s}$

b $\dfrac{32}{\pi}\ \mathrm{m/s}$